

Then, we can solve this for x , using the quadratic formula taught to teenagers everywhere:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

In this case, we find that $x = 0$ and $x = 2$ are the solutions. The first solution, $x = 0$, is where the ball started, where it was thrown by the pitcher or hit by the batter. The second solution, $x = 2$, is where the ball returns to the ground again after its flight.

The strategy we just used is a relatively simple one. Let's call it the *solve-for- x strategy*. We write down an equation that describes a situation, and then solve that equation for the variable we're interested in. The solve-for- x strategy is extremely common in the hard sciences, at both the high school and college levels. Students are asked to solve for: a ball's expected destination, the ideal level of economic production, the proportion of a chemical that should be used in an experiment, or any number of other things.

The solve-for- x strategy is extremely powerful. If, for example, an army observed an enemy force fire a projectile weapon (say, a missile), they could quickly plug Galileo's equation into their calculators and nearly instantaneously find where the missile was expected to land, and evade it or intercept it accordingly. It could be done for free on a consumer-level laptop running Python. If a robot were playing outfield in a baseball game, it could do the same to catch a ball without breaking a sweat.

The solve-for- x strategy is easy in this case because we already know the equation that needs to be solved and the method to solve it. We owe the equation for a thrown ball to Galileo, as mentioned. We owe the quadratic formula to the great Muhammad ibn Musa al-Khwarizmi, who was the first to specify a fully general solution of the quadratic equation.

Al-Khwarizmi was a ninth-century polymath who contributed to astronomy, cartography, and trigonometry, besides giving us the word *algebra* and the method it refers to. He's one of the important figures who has enabled us to take the journey of this book. Since we live after giants like Galileo and al-Khwarizmi, we don't need to suffer through the difficult